Electricity and Magnetism

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Introduction to Electricity and Magnetism

Electricity and magnetism are two closely related forces of nature. Electricity involves electric charges, which can be at rest (static electricity) or in motion (electric current). Magnetism involves the forces produced by magnets and by moving charges. In the nineteenth century scientists discovered that the two are different aspects of one force, electromagnetism, which is one of the four fundamental forces of nature. Electromagnetism powers modern civilisation: electric motors, generators, transformers, radio, television, computers and mobile phones all depend on it.

Basic Concepts of Electricity

Electric charge is a property of matter carried by protons (positive) and electrons (negative). Like charges repel and unlike charges attract. Electric current is the flow of charge in a conductor, measured in amperes (A). Voltage, or potential difference, is the energy per unit charge that pushes the current, measured in volts (V). Resistance is the opposition a material offers to current, measured in ohms. Conductors such as copper and aluminium allow current to flow easily, while insulators such as rubber and plastic do not. Electric power is the rate of using energy, P = V x I, measured in watts. Energy used in homes is measured in kilowatt-hours (kWh), also called units. Direct current (DC) flows in one direction, as in batteries, while alternating current (AC) reverses direction periodically. India's household supply is 230 volts AC at a frequency of 50 hertz.

What Is Ohm's Law?

Ohm's law states that the current flowing through a conductor is directly proportional to the voltage across it, provided the temperature and other physical conditions remain constant. The formula is V = I x R, where V is the voltage in volts, I is the current in amperes and R is the resistance in ohms. It was discovered by the German physicist Georg Simon Ohm in 1827. For example, if a 12-volt battery is connected across a 4-ohm resistor, the current is 12 divided by 4, which equals 3 amperes. If the voltage doubles, the current doubles. Ohm's law is the basic tool for analysing electric circuits. In a series circuit, components are connected end to end, the same current flows through all, and the total resistance is the sum of the individual resistances. In a parallel circuit, components are connected across the same two points, the voltage across each is the same, and the total resistance is lower than the smallest individual resistance. Household wiring is parallel so each appliance gets full voltage. Some materials, such as diodes, do not follow Ohm's law and are called non-ohmic.

Basic Concepts of Magnetism

A magnet has two poles, north and south. Like poles repel and unlike poles attract. A magnet is surrounded by a magnetic field, represented by field lines that run from north to south outside the magnet. The Earth acts like a giant magnet, which is why a compass needle points north. Materials such as iron, nickel and cobalt are ferromagnetic and can be strongly magnetised. A permanent magnet keeps its magnetism, while an electromagnet works only when current flows through a coil of wire, usually wound around an iron core, and its strength can be controlled.

The Link Between Electricity and Magnetism

In 1820 the Danish scientist Hans Christian Oersted discovered by accident that an electric current in a wire deflects a nearby compass needle. This showed that an electric current produces a magnetic field. The direction of the field can be found by the right-hand thumb rule. Soon afterwards Andre-Marie Ampere developed the mathematics of this relationship. A current-carrying wire placed in a magnetic field experiences a force, which is the principle of the electric motor. The direction of this force is found using Fleming's left-hand rule.

What Is Electromagnetic Induction?

Electromagnetic induction is the process of producing an electromotive force (voltage) and current in a conductor by changing the magnetic field around it, or magnetic flux through it. If a magnet is moved into or out of a coil of wire, a current flows in the coil while the magnet is moving. The faster the change, the greater the induced voltage. Faraday's law of induction states that the induced voltage is proportional to the rate of change of magnetic flux. Lenz's law adds that the direction of the induced current is such that it opposes the change that caused it. Fleming's right-hand rule gives the direction of induced current in a moving conductor.

Who Discovered Electromagnetic Induction?

Electromagnetic induction was discovered by the English scientist Michael Faraday in 1831. In an experiment on 29 August 1831 he wound two coils on an iron ring and found that a brief current appeared in one coil when the current in the other was switched on or off. The American scientist Joseph Henry discovered the effect independently around the same time. Later, in the 1860s, James Clerk Maxwell combined electricity, magnetism and light in four equations, predicting electromagnetic waves, which Heinrich Hertz generated in 1887.

Applications of Electromagnetic Induction

Electric generators use induction: turbines driven by steam, water, wind or nuclear heat rotate coils in magnetic fields to produce electricity in power plants. Transformers use induction to step voltage up for transmission and down for homes. Other applications are induction cooktops, wireless phone charging, metal detectors, electric guitar pickups, and the regenerative braking of electric vehicles.

Electricity and Magnetism in India

India's electricity is generated mostly from coal, along with hydro, solar, wind and nuclear sources, and distributed through a national grid. Tamil Nadu is a leader in wind power. The Indian scientist Jagadish Chandra Bose carried out pioneering experiments on millimetre-wavelength electromagnetic waves in the 1890s, shortly after Hertz. Maglev trains and magnetic resonance imaging (MRI) machines use strong magnets. Chennai's industries manufacture transformers, motors and switchgear.

Challenges and Future

Challenges include energy losses in transmission, safe handling and the integration of renewable sources. Future directions include superconducting magnets, smart grids and wireless power transfer.

Quick Facts

- Ohm's law states V = I x R: voltage equals current times resistance.
- Electromagnetic induction is the generation of voltage in a conductor by a changing magnetic field.
- Michael Faraday discovered electromagnetic induction in 1831.
- Hans Christian Oersted showed in 1820 that electric current produces a magnetic field.
- India's household supply is 230 volts AC at 50 Hz.